WORKED SOLUTION

STEP II 2013, Question 10

Mechanics • Projectile Motion

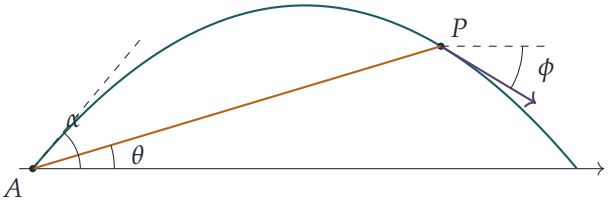
STEP II 2013, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q10

The key idea

Write down the position and velocity at time t and you'll find that $\tan\theta$ and $\tan\phi$ are both *linear* in t. The identity in (i) drops straight out, and it turns both B and C into statements about the tangents of angles. Because both angles decrease steadily with time, "which comes first" is just a comparison of two numbers, and (i) lets us make that comparison at the single point B.



At a general point P, θ is the angle of elevation of P seen from A and ϕ is the angle the velocity makes with the horizontal (negative in the picture, since the particle is falling).

Solution

We model the particle as moving freely under gravity g with no air resistance. Let it be projected with speed u at time $t = 0$, with $0 < \alpha < 90°$. Taking A as origin, at time t

$$x = ut\cos\alpha, \qquad y = ut\sin\alpha - \tfrac{1}{2}gt^2, \qquad \dot{x} = u\cos\alpha, \qquad \dot{y} = u\sin\alpha - gt.$$

It lands when $y = 0$ again, at $t = 2u\sin\alpha/g$.

Part (i)

For $t > 0$ the angle of elevation of the particle from A satisfies

$$\tan\theta = \frac{y}{x} = \frac{ut\sin\alpha - \frac{1}{2}gt^2}{ut\cos\alpha} = \tan\alpha - \frac{gt}{2u\cos\alpha},$$

and the direction of motion satisfies

$$\tan\phi = \frac{\dot{y}}{\dot{x}} = \frac{u\sin\alpha - gt}{u\cos\alpha} = \tan\alpha - \frac{gt}{u\cos\alpha}.$$

Doubling the first and using the second,

$$2\tan\theta = 2\tan\alpha - \frac{gt}{u\cos\alpha} = \tan\alpha + \left(\tan\alpha - \frac{gt}{u\cos\alpha}\right) = \tan\alpha + \tan\phi,$$

as required.

It's worth recording what we'll use later: writing $k = \dfrac{g}{u\cos\alpha} > 0$,

$$\tan\theta = \tan\alpha - \tfrac{1}{2}kt, \qquad \tan\phi = \tan\alpha - kt, \tag{$*$}$$

so both $\tan\theta$ and $\tan\phi$ strictly decrease as time goes on. Over the flight θ falls from α to 0 and ϕ falls from α to $-\alpha$, so each of $\theta = \frac{1}{2}\alpha$ and $\phi = -\frac{1}{2}\alpha$ happens exactly once: B and C are well defined.

Part (ii)

Write $T = \tan\frac{1}{2}\alpha$; since $0 < \frac{1}{2}\alpha < 45^\circ$, we have $0 < T < 1$. At B, $\tan\theta = T$. At C, $\tan\phi = \tan(-\frac{1}{2}\alpha) = -T$. If B and C are the same point, both hold there, and (i) gives

$$2T = \tan\alpha - T, \qquad \text{that is} \qquad \tan\alpha = 3T.$$

By the double-angle formula $\tan\alpha = \dfrac{2T}{1-T^2}$, so

$$\frac{2T}{1-T^2} = 3T \implies 2 = 3(1-T^2) \implies T^2 = \tfrac{1}{3},$$

where we divided by T, which is allowed because $T > 0$. Taking the positive root, $\tan\frac{1}{2}\alpha = \frac{1}{\sqrt{3}}$, so $\frac{1}{2}\alpha = 30^\circ$ and

$$\alpha = 60^\circ.$$

Part (iii)

> **Ask yourself** How can we tell which point is reached first, without working out the two times?

By $(*)$, ϕ decreases throughout the flight. So C comes *after* B exactly when, at the moment the particle is at B, ϕ has not yet fallen to $-\frac{1}{2}\alpha$; that is, when $\tan\phi_B > -T$, where ϕ_B is the direction of motion at B.

At B, part (i) gives $\tan\phi_B = 2\tan\theta - \tan\alpha = 2T - \tan\alpha$. Hence

$$B \text{ is reached first} \iff 2T - \tan\alpha > -T \iff \tan\alpha < 3T.$$

Using $\tan\alpha = \dfrac{2T}{1-T^2}$ again, and dividing by $T > 0$ and multiplying by $1 - T^2 > 0$ (which keeps the inequality's direction),

$$\frac{2T}{1-T^2} < 3T \iff 2 < 3 - 3T^2 \iff T^2 < \tfrac{1}{3} \iff \tfrac{1}{2}\alpha < 30^\circ,$$

the last step because tan is increasing on $(0^\circ, 45^\circ)$. Every step is reversible, so

$$\alpha < 60^\circ \iff B \text{ is reached before } C.$$

So, given $\alpha < 60^\circ$, the particle reaches B first.

If you prefer the times explicitly: from $(*)$, $kt_B = 2(\tan\alpha - T)$ and $kt_C = \tan\alpha + T$, and $t_B < t_C$ is again equivalent to $\tan\alpha < 3T$.

Checks

- At $t = 0$ both formulas in $(*)$ give $\tan\alpha$, as they should: just after launch the particle is moving along the line of projection. At landing, $kt = 2\tan\alpha$, giving $\tan\theta = 0$ and $\tan\phi = -\tan\alpha$, the familiar symmetry of the trajectory.
- At $\alpha = 60^\circ$, $\tan\alpha = \sqrt{3} = 3\tan 30^\circ$, consistent with (ii).
- With $u = 10\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $\alpha = 45^\circ$, the formulas give $t_B \approx 0.845\,\mathrm{s}$ and $t_C \approx 1.020\,\mathrm{s}$, so B comes first; stepping the motion numerically by computer gives the same times, and also confirms that C comes first for $\alpha = 61^\circ$, 75° and 85°.

Common slips

- Treating θ (the direction of the particle *from* A) as if it were ϕ (the direction of its *velocity*). They agree only at the instant of launch.
- Getting the sign wrong at C: there $\tan\phi = -\tan\frac{1}{2}\alpha$, not $+\tan\frac{1}{2}\alpha$.
- In (iii), claiming the answer without saying why "$\tan\phi_B > -T$" means B comes first. It's the monotonic decrease of ϕ that justifies it.

Beyond STEP

The identity in (i) says that the chord from A to any point of a parabola has a slope equal to the average of the slopes at its two ends, a special property of quadratics (it's the mean value theorem with the midpoint pinned down exactly). In a first-year mechanics course you'd meet the same ideas when studying projectiles on inclined planes, where the angle of the line from the launch point plays exactly the role of θ here. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see projectile problems treated with this kind of care.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.